Strand 3: Geometry and Measurement

Sub-strand 1: Shapes and Space

Lesson 4 (Requested topic): Working with Different Types of Triangles

1. Curriculum Standard and Overview

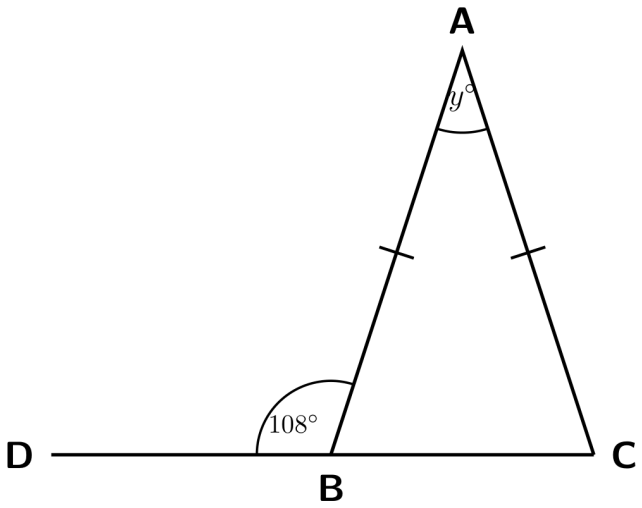
- Strand: 3 — Geometry and Measurement
- Sub-strand: 1 — Shapes and Space
- Content Standards: B8.3.1.1 (properties of shapes and space), B8.3.2.1 (measurement with right-angled triangles).
- Indicators: B8.3.1.1.2 (determine angles in a triangle using the angle-sum fact), B8.3.2.1.2 (relationship between hypotenuse c and sides a, b — Pythagoras), B8.3.2.1.3 (solve right-triangle problems), B8.3.2.1.4 (use Pythagoras in real-life problems).
- Why this lesson before the others: the area of a triangle (next lesson) and the construction of triangles (Constructions I & II) both require you to know the three side-families, the three angle-families, the angle sum, and Pythagoras' theorem. This is the foundation lesson.

Already delivered and assumed: angle facts (straight line 180°, point 360°, vertically opposite, parallel lines) from Lesson 3 of this folder; triangle = polygon with n = 3, so angle sum = (3-2)×180 = 180° from Lesson 1 (Polygons).

2. Preamble — Six Names for Any Triangle

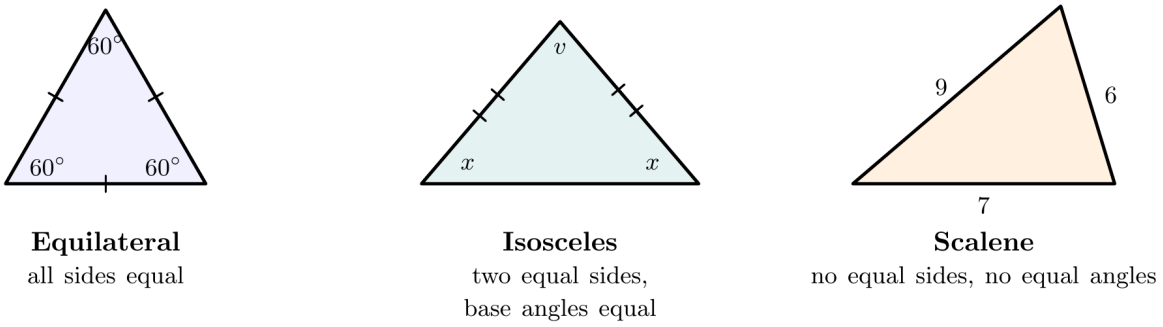
Every triangle gets two names at once:

1. A name by its sides (how many sides are equal).
2. A name by its largest angle (how wide the widest corner is).



By sides	Meaning	Angle fact
Equilateral	all 3 sides equal	all 3 angles = 60°
Isosceles	exactly 2 sides equal	the 2 base angles (opposite the equal sides) are equal
Scalene	no sides equal	no angle equal

By largest angle	Meaning
Acute-angled	all angles < 90°
Right-angled	exactly one angle = 90°
Obtuse-angled	one angle between 90° and 180°



Equilateral
all sides equal

Isosceles
two equal sides,
base angles equal

Scalene
no equal sides, no equal angles

Figure 3 shows the three side names side by side; equal ticks mark equal sides, and the letters x, x mark the equal base angles of the isosceles.

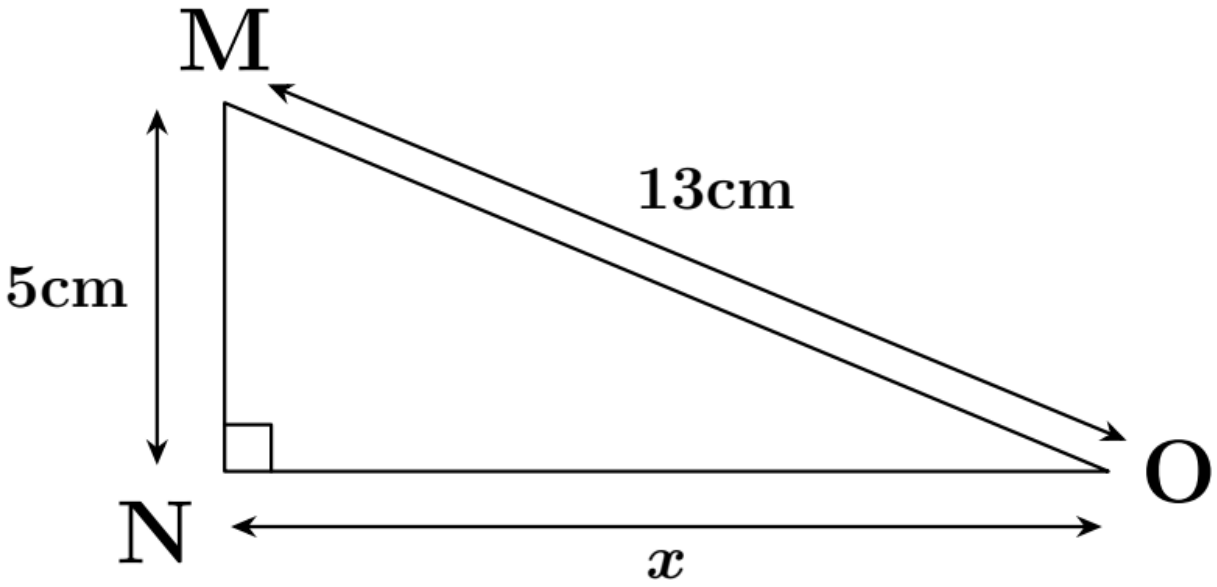
Example: a triangle with sides 5, 5, 8 and all angles below 90° is isosceles acute-angled. A triangle with angles 90°, 30°, 60° is scalene right-angled.

2.1 The three angle facts inside every triangle

1. Angle sum: angle A + angle B + angle C = 180°.
2. Isosceles base angles: the angles opposite the equal sides are equal.
3. Exterior angle: an exterior angle equals the sum of the two opposite (remote) interior angles.

In Figure 1, the exterior angle at the extended vertex = the two far interior angles added.

3. Core Concept — Pythagoras' Theorem (the right-triangle engine)



In any right-angled triangle, with c the hypotenuse (the longest side, opposite the right angle) and a, b the two shorter sides:

$$a^2 + b^2 = c^2$$

Three uses:

Find	Rearrangement
hypotenuse c	c = √(a^(2)+b^(2))
short side	a = √(c^(2) - b^(2)) (subtract, never add!)

Converse test (to classify a triangle): given three sides, check the biggest side c:

- a^(2) + b^(2) = c^(2) → right-angled
- a^(2) + b^(2) > c^(2) → acute-angled
- a^(2) + b^(2) < c^(2) → obtuse-angled

Must-know Pythagorean triples: (3,4,5), (5,12,13), (8,15,17), (7,24,25) — and their multiples like (6,8,10), (9,12,15), (10,24,26).

4. Fifteen Fully Worked Examples

Example 1 — Classification check (own)

A triangle has angles 55°, 45°, 80°. Name it fully.

All angles < 90° → acute-angled; all three angle values differ (so no equal opposite sides) → scalene. Name: acute-angled scalene triangle. (Check: 55+45+80 = 180 .)

Example 2 — BECE 2003 (authentic)

In triangle XYZ, angle XZY = 90°, |XY| = 13 cm and |YZ| = 5 cm. Find |XZ|.

|XY| = 13 is opposite the right angle → hypotenuse. |XZ|^(2) = 13^(2) - 5^(2) = 169 - 25 = 144, so |XZ| = 12 cm. (Triple 5-12-13.)

Example 3 — BECE 2015 (authentic)

The hypotenuse and a side of a right-angled triangle are 13 cm and 5 cm. Find the third side.

Same triple: third side = √(13^(2)-5^(2)) = √(144) = 12 cm.

Example 4 — Isosceles vertex angle (own)

An isosceles triangle has vertex angle 40°. Find each base angle.

Base angles = (180 - 40) ÷ 2 = 140 ÷ 2 = 70° each.

Example 5 — BECE 1997 (authentic)

In triangle PQR, |PQ| = |QR| and angle PQR = 90°. Find the equal base angles.

|PQ|=|QR| → equal angles at R and P. 2x + 90 = 180 → x = 45° each. (This is the famous isosceles right triangle — halves of a square.)

Example 6 — Exterior angle (own)

Two interior angles of a triangle are 55° and 65°. Find the exterior angle at the third vertex.

Exterior angle = 55 + 65 = 120°. (Check: interior at that vertex = 180-55-65 = 60°, and its straight-line partner is 180 - 60 = 120 .)

Example 7 — BECE 1998 (authentic)

PQR is isosceles with |PQ| = |PR|, angle QPR = 40°, and QRS is a straight line. Find angle PRS.

Base angles: (180-40)÷2 = 70° each, so angle PRQ = 70°. angle PRS is on the straight line QRS: angle PRS = 180 - 70 = 110°. (Shortcut: exterior angle = sum of remote interiors = 70 + 40 = 110 , matches the authentic key.)

Example 8 — Algebra angles (own)

The angles of a triangle are x, 2x, 3x. Find x and name the triangle.

6x = 180 → x = 30. Angles: 30°, 60°, 90° → right-angled scalene.

Example 9 — Isosceles + algebra (own)

An isosceles triangle has base angles x each and vertex angle 4x. Find all angles and name it.

6x = 180 → x = 30. Angles: 30°, 30°, 120° → obtuse-angled isosceles.

Example 10 — BECE 2002 (authentic, area × right angle)

PQR is a right-angled triangle. Its area is 6 cm^2 and |QR| = 3 cm. Find the other short side (the printed options are 2, 3, 4, 6 — the correct maths gives one of these).

With the right angle between the two short sides, area = ½ × 3 × h = 6 → h = 4 cm. So the other short side is 4 cm. (Note: if that side's name in your printed paper differs, match the side by the working — the value is 4 cm.)

Example 11 — BECE 2002 (authentic — Pythagoras inside a rectangle)

A rectangle has length 8 cm and breadth 6 cm. How long is its diagonal?

The diagonal cuts the rectangle into two right triangles with legs 8, 6: d^(2) = 8^(2) + 6^(2) = 64 + 36 = 100 → d = 10 cm. (Triple 6-8-10.)

Example 12 — Converse test (own)

Classify the triangles: (a) sides 6, 8, 10; (b) sides 4, 5, 6.

(a) 6^(2) + 8^(2) = 100 = 10^(2) → right-angled scalene. (b) biggest side 6: 4^(2)+5^(2) = 41 > 36 = 6^(2) → acute-angled scalene.

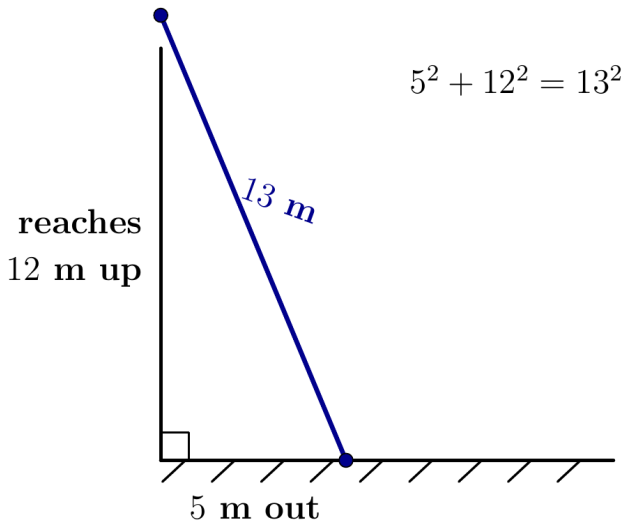
Example 13 — BECE 1994 style (height of an isosceles)

Triangle ABC: |AB| = |BC| = 5 cm, |AC| = 8 cm. BD is the perpendicular from B to AC. Find |BD|.

D is the midpoint of AC (isosceles symmetry): |AD| = 4 cm. |BD|^(2) = 5^(2) - 4^(2) = 9 → |BD| = 3 cm.

Example 14 — Real-life (own)

A ladder 13 m long leans against a wall with its foot 5 m out on level ground. How high does it reach?



Wall = √(13^(2) - 5^(2)) = √(144) = 12 m.

Example 15 — Equilateral fact (own, P2 style)

An equilateral triangle has side 6 cm and vertical height 5.2 cm. Find (a) each angle, (b) its area.

(a) Each angle = 60°. (b) A = ½ b h = ½ × 6 × 5.2 = 15.6 cm^2.

5. Exercises — 5 Sections × 5 Questions (No MCQ)

Show every step. State the type of triangle when asked.

Exercise 1 — Naming triangles

1. A triangle has angles 100°, 40°, 40°. Name it fully (both names).
2. A triangle has sides 7 cm, 7 cm, 7 cm. Name it and state all three angles.
3. A triangle has sides 4, 6, 8. Use the converse test to name it.
4. A triangle has sides 9, 12, 15. Name it.
5. Give one word meaning: a triangle with exactly two equal angles.

Exercise 2 — Angle sum and isosceles work

1. Two angles of a triangle are 48° and 61°. Find the third.
2. An isosceles triangle has base angles of 64° each. Find the vertex angle.
3. An isosceles triangle has vertex angle 96°. Find each base angle.
4. The angles of a triangle are in the ratio 2 : 3 : 4. Find them all and name the triangle.
5. In an isosceles triangle the vertex angle is (3x)° and each base angle is (x + 15)°. Find x and all angles.

Exercise 3 — Exterior angles

1. Interior angles of 72° and 58°; find the exterior angle at the third vertex.
2. An exterior angle of a triangle is 135° and one remote interior angle is 85°. Find the other remote interior angle, and the interior angle at the same vertex.
3. triangle ABC is isosceles, |AB| = |AC|, vertex angle A = 36°. The side BA is extended to D. Find angle CAD.
4. In triangle PQR, angle PQR = 70° and angle QRP = 65°. Side QR is produced to S. Find angle PRS.
5. One exterior angle of a triangle is 110°; an adjacent interior angle (at the same vertex) is x° and a different interior angle is 55°. Find x and the third angle.

Exercise 4 — Pythagoras engine

1. Legs 9 cm and 12 cm: find the hypotenuse.
2. Hypotenuse 17 cm, one leg 8 cm: find the other leg.
3. A rectangle is 24 cm by 7 cm. Find its diagonal.
4. Triangle sides 5, 12, 13: prove it is right-angled, then find its area (right angle between the two shorter sides).
5. A sail is a right triangle: two edges of 15 cm and 20 cm meet at a right angle. How long is the closing (third) edge?

Exercise 5 — Spiral review

1. (Perimeter & area, Lesson 3) A square has perimeter 96 cm; find its area.
2. (Polygons, earlier geometry) A regular pentagon has interior angle 108° each. Verify with the formula (n-2)×180 ÷ n.
3. (Algebra) Solve 5(x-2) = 3x + 6 and use the answer as an angle in degrees; state whether it is acute, right, obtuse or reflex.
4. (Percent) A triangle's area is 150 cm^2. A sticker covers 40% of it. Find the sticker area.

5. (Congruence, earlier geometry) triangle ABC cong triangle DEF with |AB| = 7 cm, angle A = 50°. State |DE| and angle D.

6. Full Worked Solutions — Exercises

Exercise 1

1. Angles 100° (obtuse, the largest), two equal 40° → obtuse-angled isosceles.
2. All sides equal → equilateral; all angles 60°.
3. Biggest side 8: 4^(2)+6^(2) = 52 < 64 = 8^(2) → obtuse-angled scalene.
4. 9^(2)+12^(2) = 81+144 = 225 = 15^(2) → right-angled scalene. (Triple 3-4-5 scaled ×3.)
5. Isosceles (equal angles opposite equal sides; the two equal angles make it so).

Exercise 2

1. 180 - (48+61) = 71°.
2. 180 - 2(64) = 52°.
3. (180-96)÷2 = 42° each.
4. Total parts 9; one part = 20°; angles 40°, 60°, 80° → acute-angled scalene.
5. 3x + 2(x+15) = 180 → 5x + 30 = 180 → x = 30. Angles: 90° vertex, 45°, 45° base → right-angled isosceles.

Exercise 3

1. 72 + 58 = 130°.
2. Other remote interior = 135 - 85 = 50°; interior at the vertex = 180 - 135 = 45°.
3. Base angles = (180-36)÷2 = 72°; angle CAD = 180 - 72 = 108° (or exterior rule 36 + 72).
4. angle PRS = 70 + 65 = 135°.
5. x = 180 - 110 = 70°; third angle = 180 - (70 + 55) = 55°. (Check with the exterior rule: remote interiors 55° + 55° = 110° .)

Exercise 4

1. c = √(81+144) = √(225) = 15 cm.
2. b = √(17^(2) - 8^(2)) = √(225) = 15 cm.
3. d = √(24^(2)+7^(2)) = √(625) = 25 cm.
4. 5^(2) + 12^(2) = 169 = 13^(2) right-angled; area = ½×5×12 = 30 units^2.
5. 15^(2) + 20^(2) = 225 + 400 = 625 = 25^(2) → string = 25 cm.

Exercise 5

1. Side = 24; area = 576 cm^2.
2. (5-2)×180÷5 = 540÷5 = 108° .
3. 5x - 10 = 3x + 6 → 2x = 16 → x = 8 → 8°, acute.
4. 0.40 × 150 = 60 cm^2.
5. |DE| = 7 cm, angle D = 50° (congruent = identical).

7. Memory Card

- Every triangle: two names (sides + angles).
- Angle sum = 180°; exterior = sum of the two remote interiors.
- Isosceles → base angles equal; equilateral → three 60° angles.
- Pythagoras: a^(2) + b^(2) = c^(2); triples (3,4,5), (5,12,13), (8,15,17), (7,24,25).
- Converse test sorts any three sides into right / acute / obtuse.

Next lesson: area of polygons — the triangle area formula ½bh is the centrepiece, and you now know how to find h inside isosceles and right triangles.